LumoHome

System Setup Guide

Version 1.0 • March 2026

PART 1: Creating a HiveMQ Cloud MQTT Broker

HiveMQ Cloud is a free cloud MQTT broker that LumoHome uses for communication between devices and the mobile app. In this section, you will create an account, launch a serverless cluster, and configure access credentials.

1.1 HiveMQ Account Registration

Step 1: Open your browser and go to: hivemq.cloud

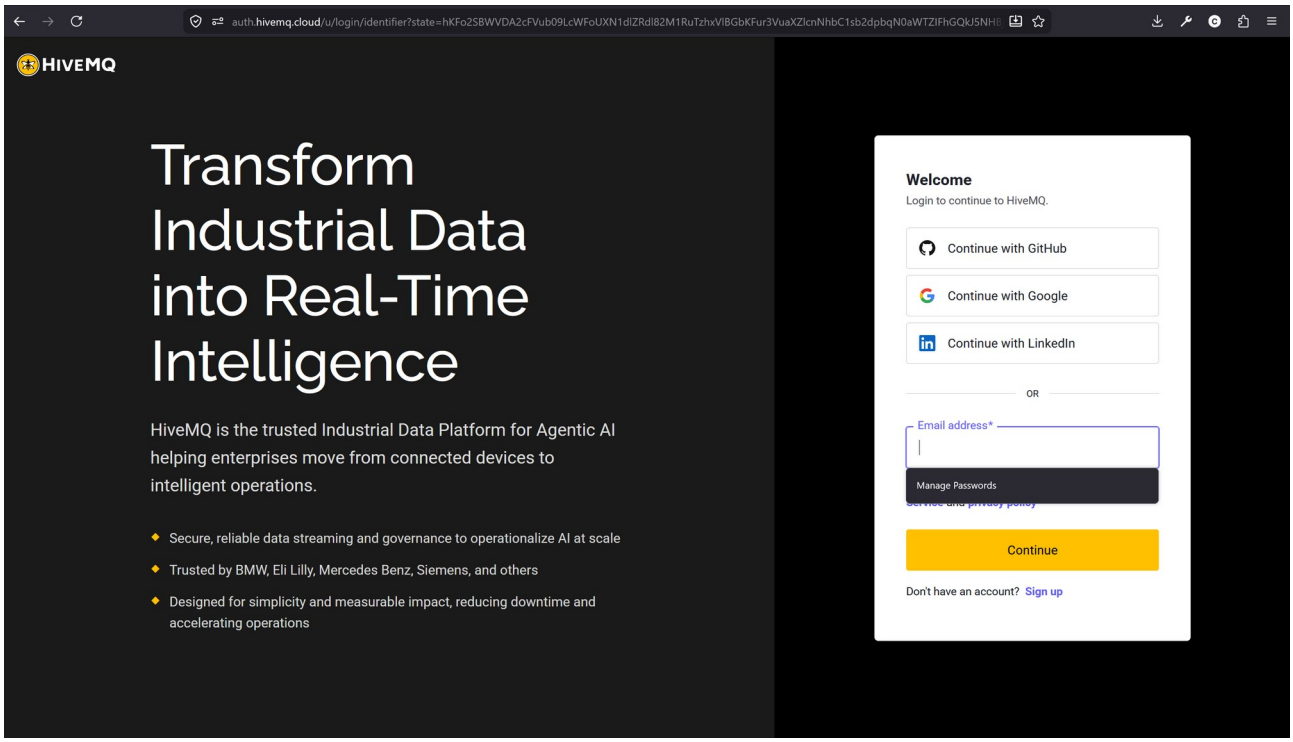

Figure 1.1 — HiveMQ Cloud login page

Step 2: Click the "Continue with Google" button and select your Google account

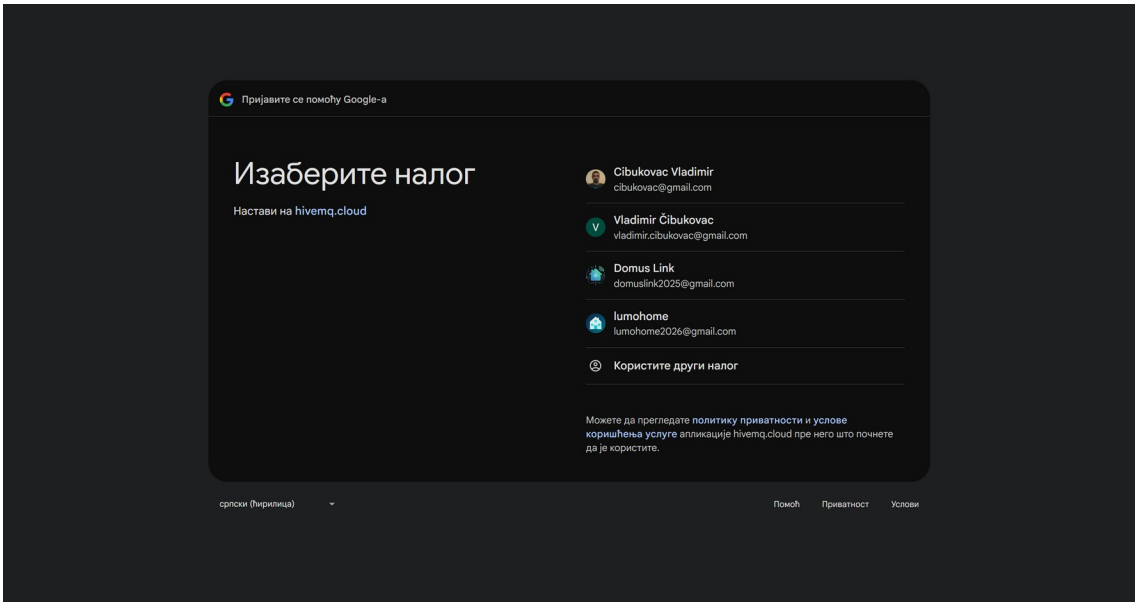

Figure 1.2 — Google account selection

Step 3: Confirm access by clicking "Continue" in the Google dialog

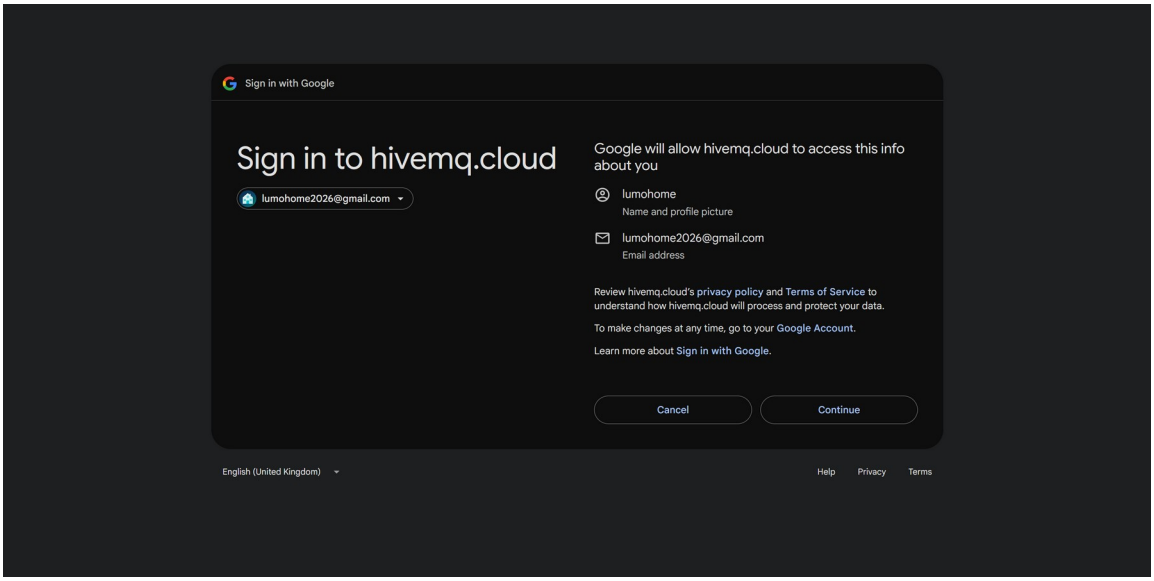

Figure 1.3 — Google login confirmation

Step 4: Enter your first and last name and click "Continue"

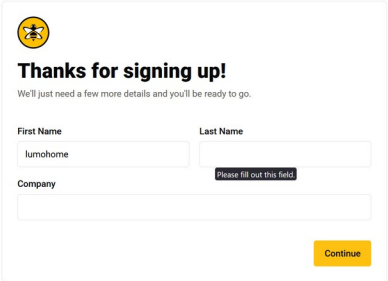

Figure 1.4 — Filling in the profile

Step 5: Accept the terms of service by clicking "I Accept the Terms of Service"

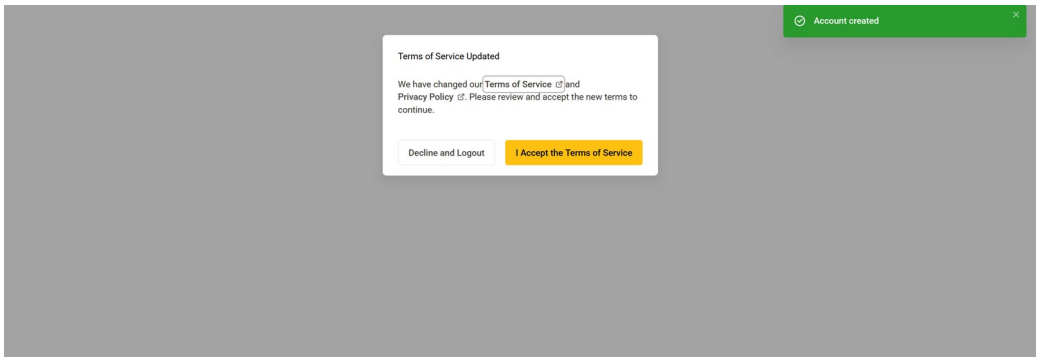

Figure 1.5 — Accepting the terms of service (account created ✓)

Step 6: You will be redirected to the HiveMQ dashboard

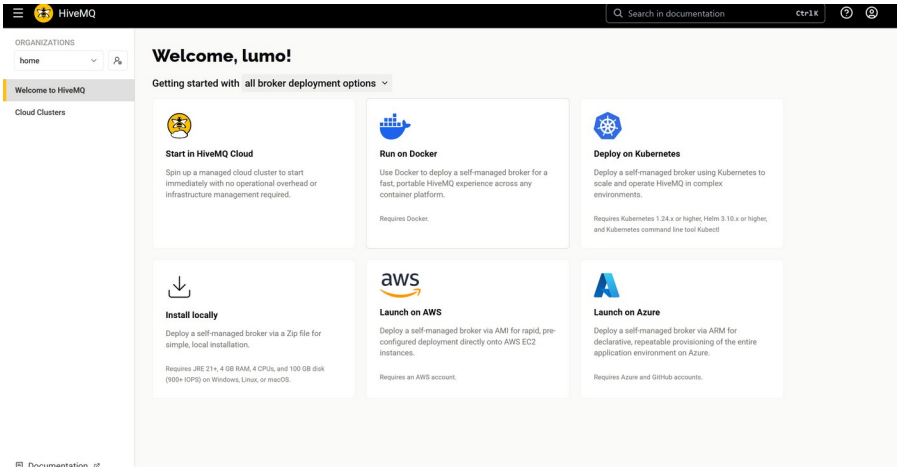

Figure 1.6 — HiveMQ dashboard (Welcome Dashboard)

1.2 Creating a Free Serverless Cluster

Step 1: In the left menu, click "Cloud Clusters"

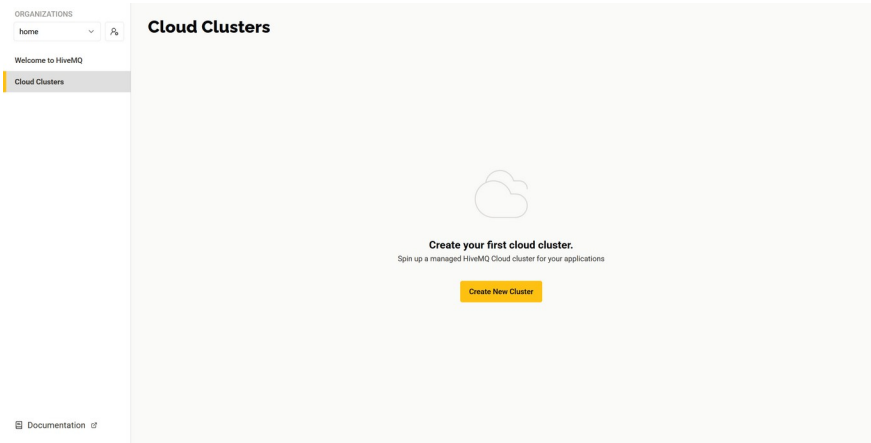

Figure 1.7 — Empty cluster list

Step 2: Click "Create New Cluster" and select the "Serverless" plan (free)

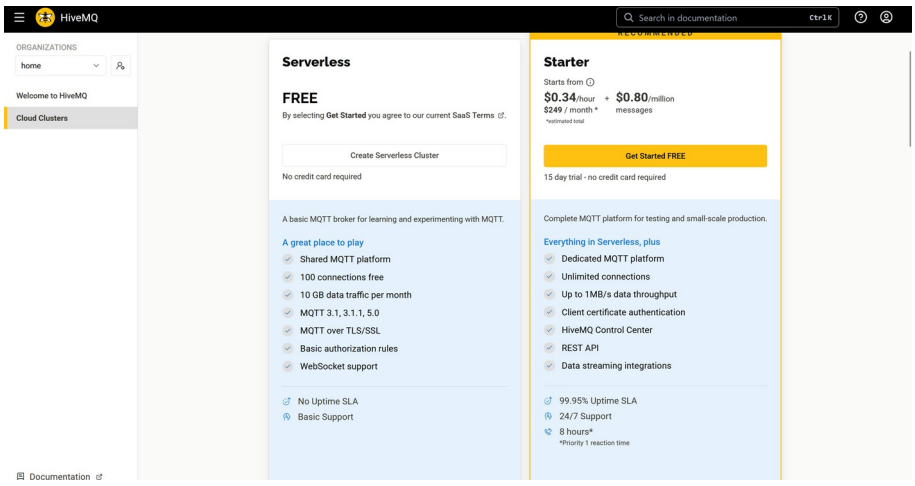

Figure 1.8 — Selecting the Serverless (FREE) plan — no credit card required

> 💡 The Serverless plan is completely free: 100 connections, 10 GB of traffic per month, supports MQTT over TLS/SSL — enough for the entire LumoHome system.

Step 3: The cluster will be up and running within a few seconds (status: Running ▲)

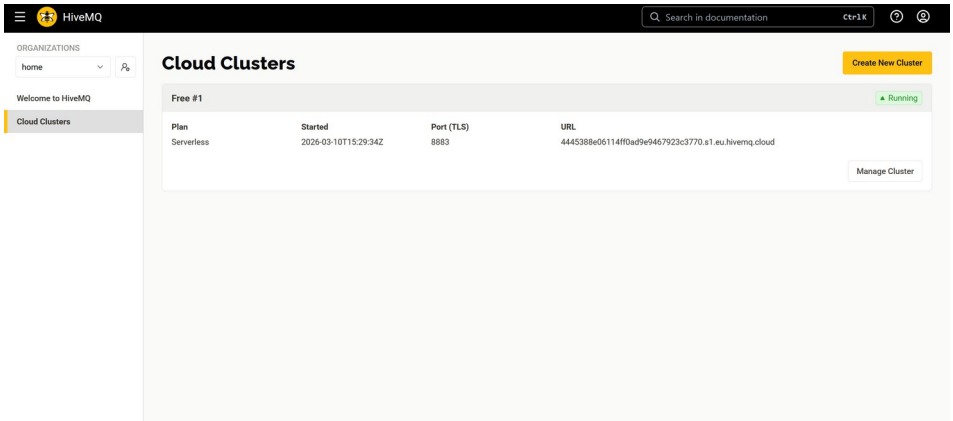

Figure 1.9 — Cluster "Free #1" in Running status

Step 4: Click "Manage Cluster" to view connection details

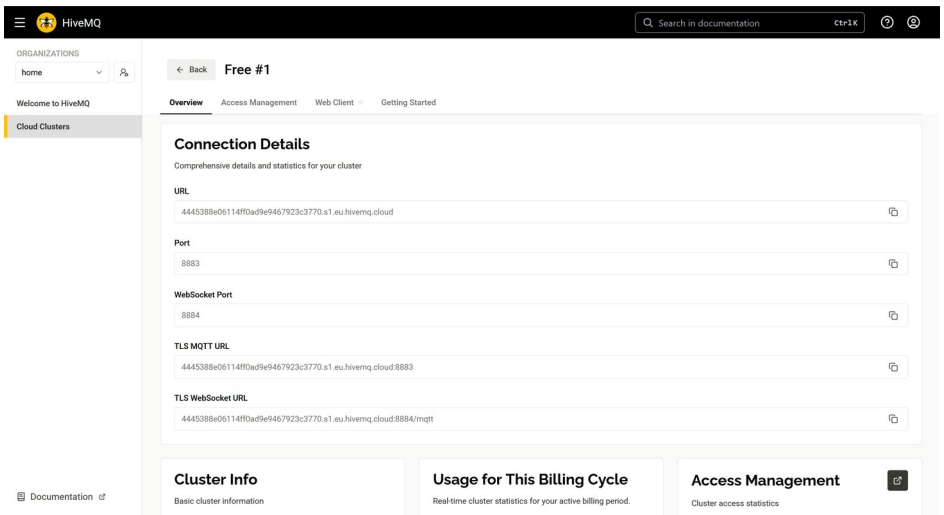

Figure 1.10 — Connection Details: URL and Port for Tasmota configuration

> 💡 Write down or copy the URL and Port (8883) — you will need them in Part 2 when configuring the Tasmota device.

1.3 Creating MQTT Credentials

Step 1: Click the "Access Management" tab

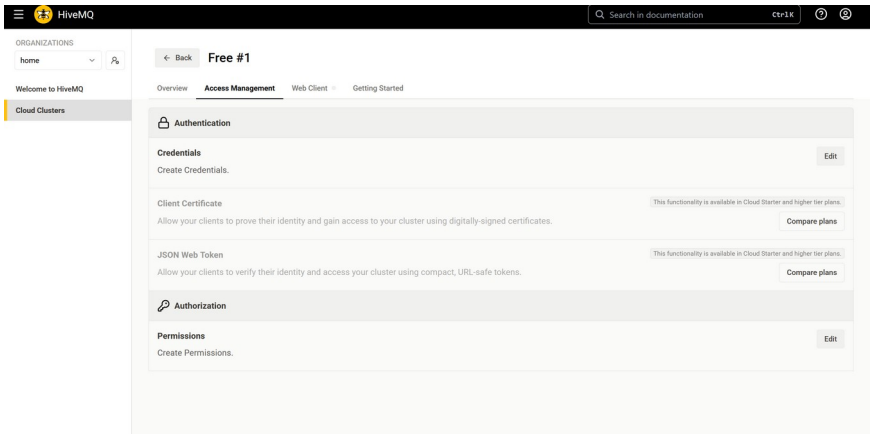

Figure 1.11 — Access management section (Authentication)

Step 2: Click "Add Credentials" or "Edit" next to Credentials

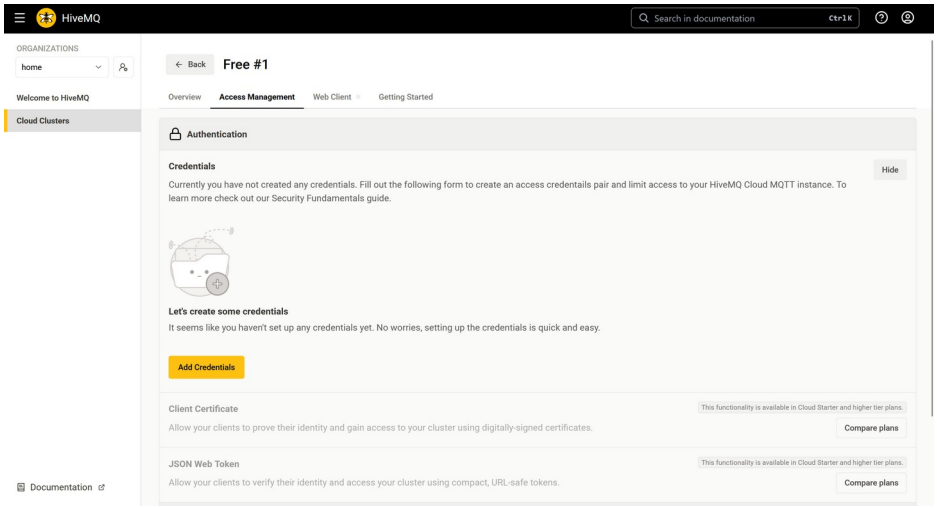

Figure 1.12 — Credential creation form

Step 3: Enter a username and password, set "Publish and Subscribe", click Save

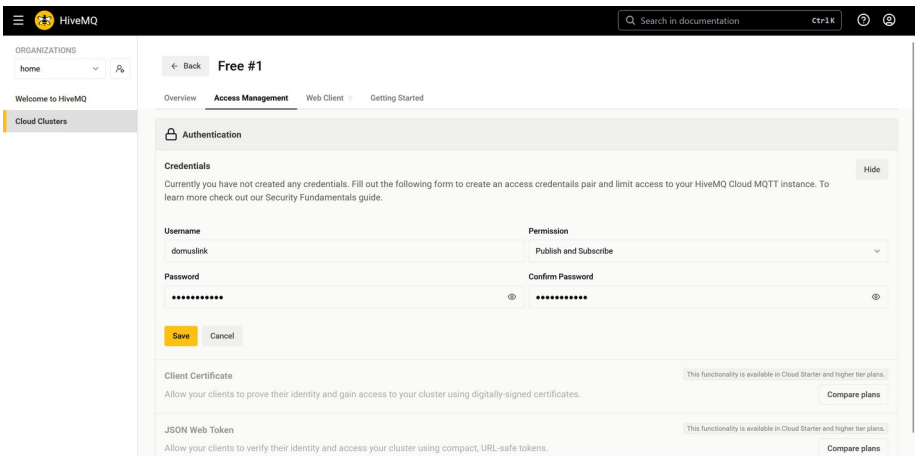

Figure 1.13 — Entering username (e.g. domuslink) and password

Step 4: Credentials are saved and active (status: Active)

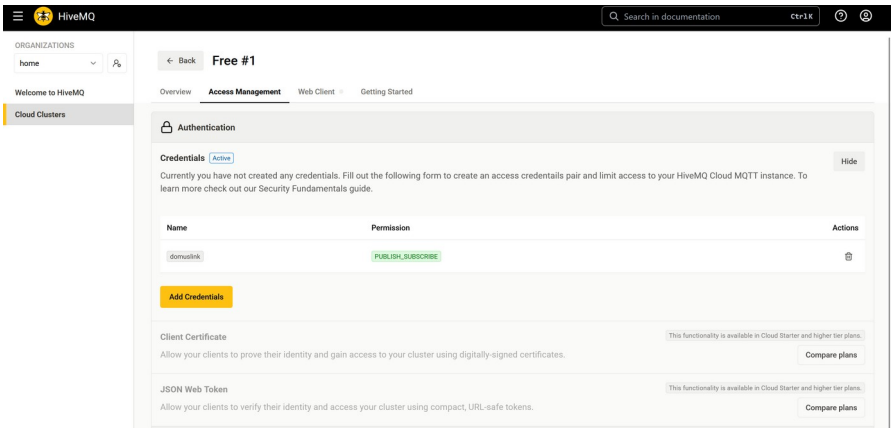

Figure 1.14 — Credentials active: domuslink / PUBLISH_SUBSCRIBE

💡 The username and password you entered will be used in the next step when configuring the Tasmota MQTT parameters.

PART 2: Configuring the Tasmota Zigbee Bridge

The Tasmota Zigbee bridge is an ESP32 device with a Zigbee chip that communicates with sensors and switches. You need to configure the WiFi network, MQTT broker, and upload the Berry script (autoexec.be).

Recommended Zigbee Bridges

LumoHome works with any Tasmota-based Zigbee bridge. Below are the tested and recommended models:

Model	Price	Notes
Athom ZG01	~25$	⭐ Recommended. ESP32 + CC2652P2, Ethernet port, Tasmota pre-installed — ready to use out of the box.
Sonoff ZBBridge-P	~15–20$	⭐ Recommended. ESP32 + CC2652P2. WiFi only, no Ethernet. Requires Tasmota flashing — for advanced users.
ZB-GW03 / EACHEN	~25–30$	WiFi + Ethernet. Good range. Pre-installed but uses an older Zigbee controller.
Tube's Gateways	~40$+	DIY/enthusiast option. See: github.com/tube0013/tube_gateways

2.1 Initial WiFi Setup

When the device is in factory state (or after a reset), it broadcasts its own WiFi network named tasmota-XXXXXX.

> **Step 1:** On your computer or phone, open WiFi settings and find the network "tasmota-XXXXXX"

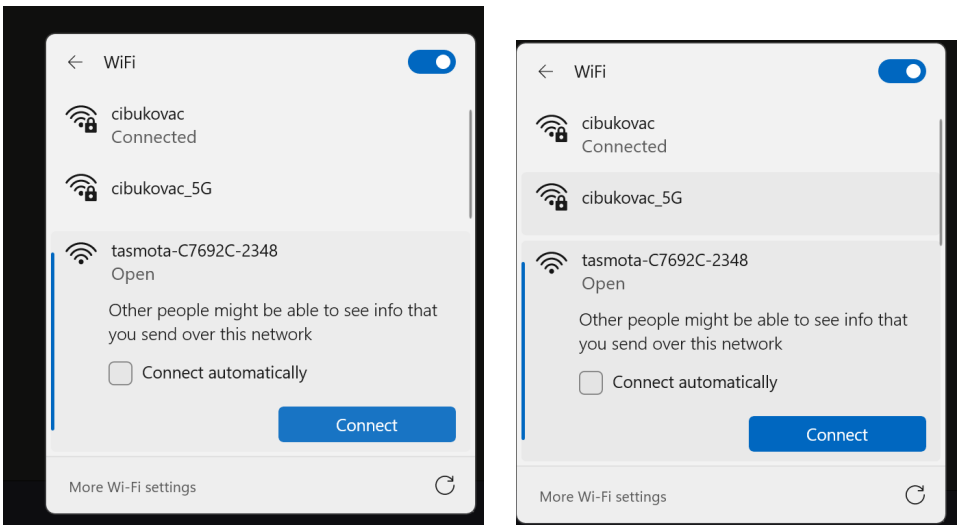

Figure 2.1 — Finding and connecting to the Tasmota hotspot (Open network)

Step 2: Connect to that network — your browser will automatically open 192.168.4.1

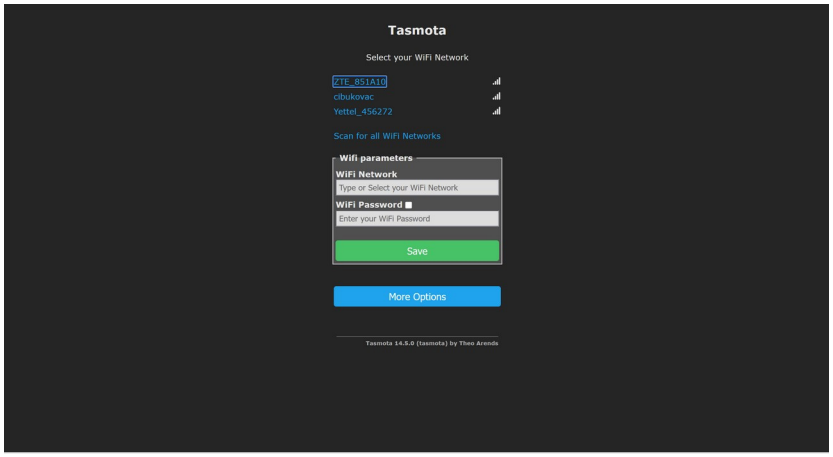

Figure 2.2 — Tasmota WiFi page: select your home network and enter the password

💡 Click on your home WiFi network name (e.g. cibukovac) to fill in the field. Enter the password and click "Save".

Step 3: After a successful connection, you will receive the new IP address of the device

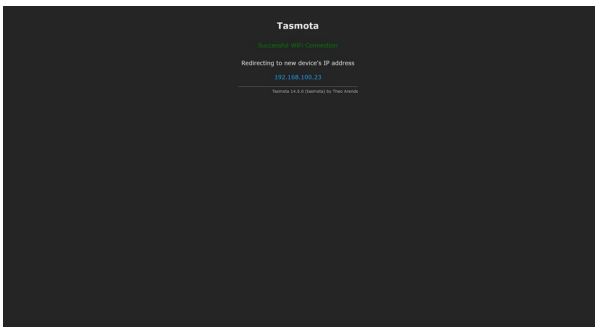

Figure 2.3 — Successful WiFi connection — new device IP address

The device now redirects to its new address on your local network (e.g. 192.168.100.23). Reconnect your computer to your home WiFi network and open that address in your browser.

2.2 MQTT Configuration

Step 1: Open the Tasmota device IP address in your browser — you will see the main screen

Figure 2.4 — Tasmota main screen (Zigbee bridge ZB-GW03)

Step 2: Click "Configuration" and then "MQTT"

Figure 2.5 — Tasmota Configuration menu

Step 3: Fill in the MQTT parameters with data from the HiveMQ cluster (Part 1)

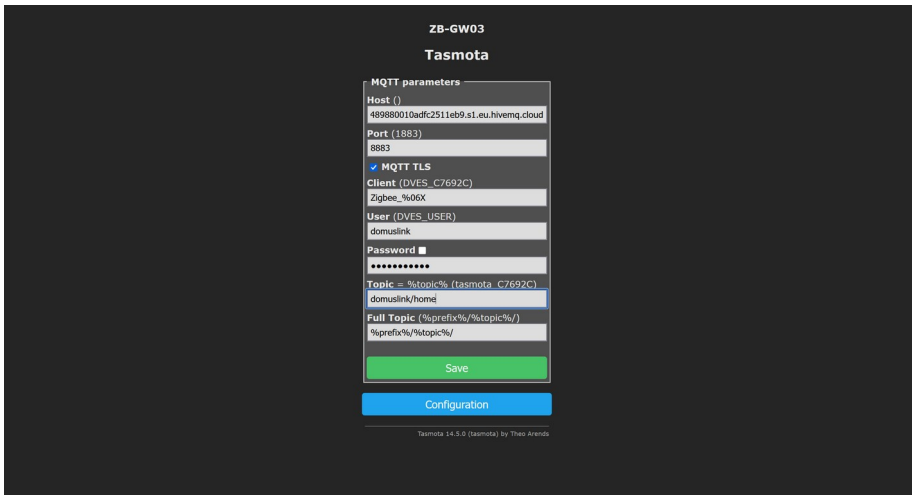

Figure 2.6 — MQTT parameters filled with HiveMQ data

Field	Value	Note
Host	`xxxx.s1.eu.hivemq.cloud`	URL from HiveMQ Connection Details
Port	`8883`	TLS port — MQTT TLS must be enabled
MQTT TLS	☑️ `(enabled)`	Without TLS the connection will not be encrypted
Client	`Zigbee_%06X`	Unique device ID (automatic)
User	`domuslink`	Username from HiveMQ Access Mgmt
Password	`••••••••`	Password from HiveMQ Access Mgmt
Topic	`domuslink/home`	MQTT topic — must match LumoHome
Full Topic	`%prefix%/%topic%/`	Leave as default

Step 4: Click "Save" — Tasmota will restart and connect to HiveMQ

Step 5: Check the console to verify that the MQTT connection has been established

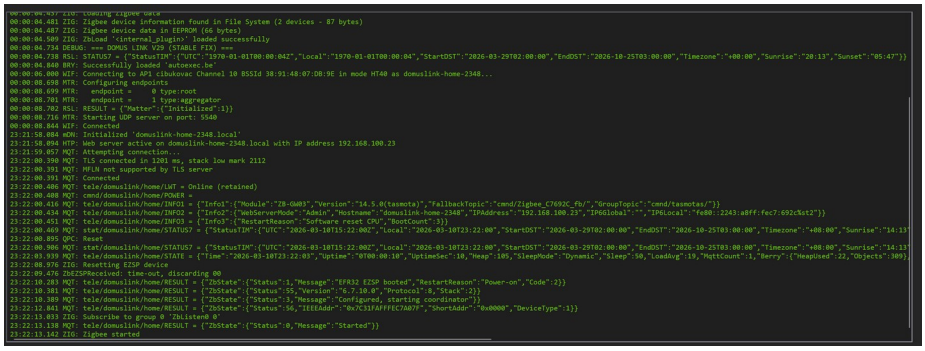

Figure 2.7 — Tasmota console: MQT: Connected ✓, ZIG: Zigbee started ✓

> 💡 Look for the following messages in the console: "MQT: Connected" and "ZIG: Zigbee started". If you see them, everything is configured correctly.

2.3 Uploading the autoexec.be Script

The autoexec.be script contains Berry code that runs on the ESP32 and manages the alarm system, automations, and RF gateway functions. It must be uploaded to the Tasmota file system.

Step 1: In the Tasmota menu, go to Tools → Manage File System

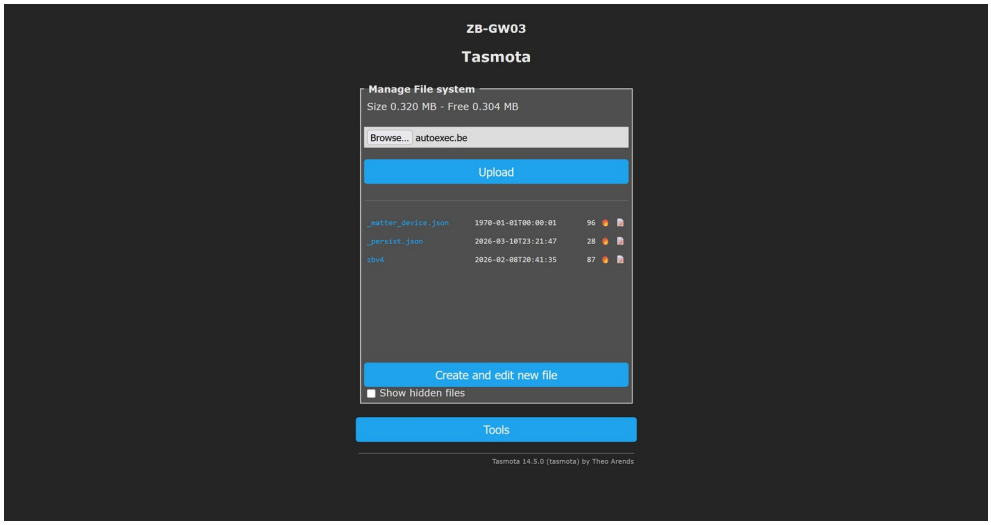

Figure 2.8 — File Manager before upload (no autoexec.be)

Step 2: Click "Browse", select the autoexec.be file and click "Upload"

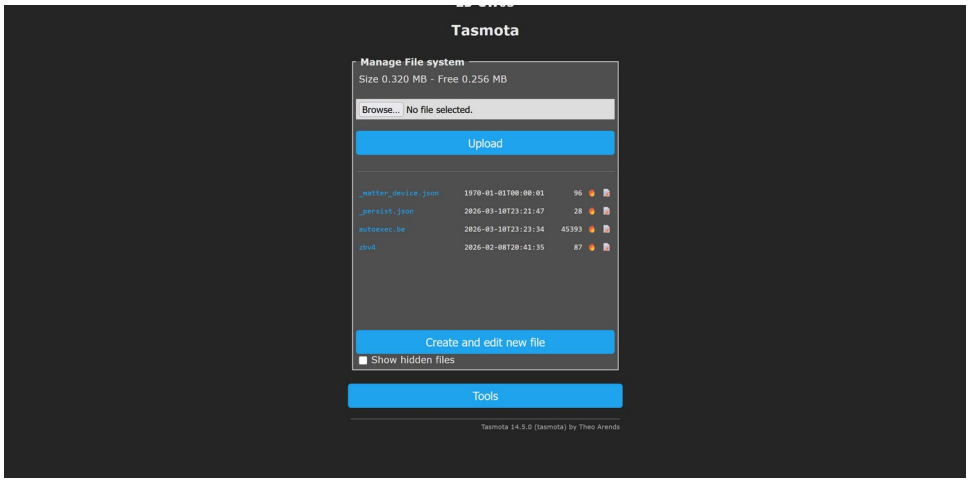

Figure 2.9 — File Manager after upload: autoexec.be (45393 bytes) ✓

💡 The autoexec.be file is included with the LumoHome app or provided by the seller. The file size is approximately 44–46 KB.

Step 3: After uploading, restart the device using the console command

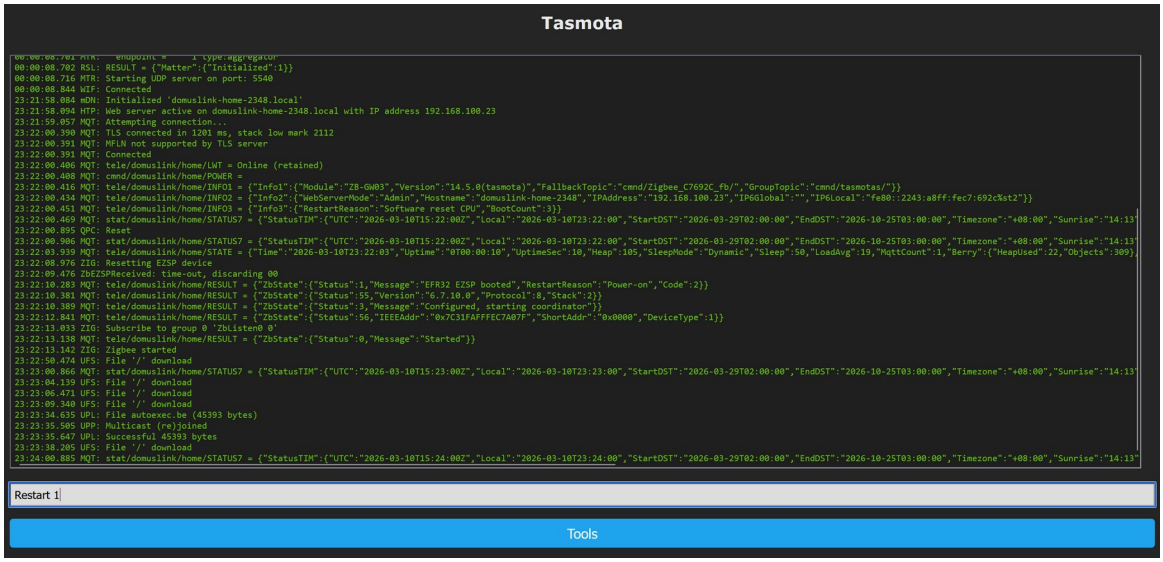

Figure 2.10 — Console: entering the "Restart 1" command to restart the device

```
Restart 1
```

2.4 Final Configuration — Backlog Command

After the restart, enter the following Backlog command in the Tasmota console to configure all remaining parameters (message retention, keepalive, time zone, NTP server):

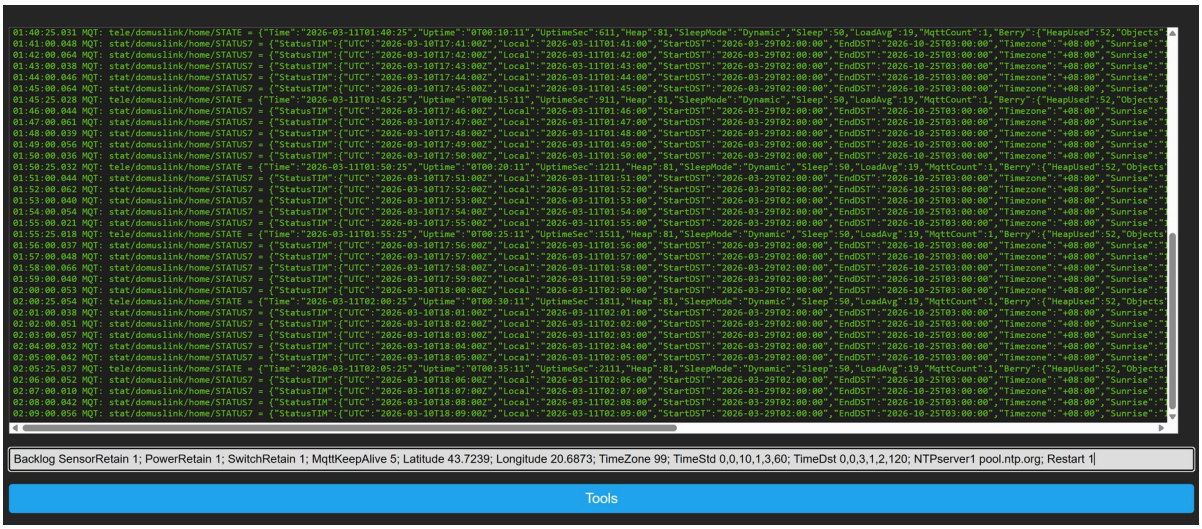

Figure 2.11 — Tasmota console with Backlog command for final configuration

Copy and paste the following command into the console field:

```
Backlog SensorRetain 1; PowerRetain 1; SwitchRetain 1; MqttKeepAlive 5;
Latitude 43.7239; Longitude 20.6873; TimeZone 99; TimeStd 0,0,10,1,3,60;
TimeDst 0,0,3,1,2,120; NTPserver1 pool.ntp.org; Restart 1
```

💡 Replace `Latitude 43.7239; Longitude 20.6873` with your own geographic coordinates. You can look them up at maps.google.com by right-clicking your location. The values shown are an example for Central Serbia.

Parameter	Description
`SensorRetain 1`	Store the last sensor value on the MQTT broker
`PowerRetain 1`	Store the power state (switches)
`SwitchRetain 1`	Store the state of physical switches
`MqttKeepAlive 5`	MQTT keepalive interval (5 seconds)
`Latitude / Longitude`	Geographic location for accurate sunrise/sunset
`TimeZone 99 + TimeStd/TimeDst`	Automatic daylight saving/standard time correction
`NTPserver1 pool.ntp.org`	Server for accurate time synchronization
`Restart 1`	Restart the device to apply the changes

PART 3: Verifying System Operation

After completing all steps, the system should be operating correctly. Check the following items:

✅	HiveMQ cluster is in "Running" status at hivemq.cloud
✅	Tasmota device is connected to the home WiFi network
✅	The Tasmota console shows the message "MQT: Connected"
✅	The Tasmota console shows the message "ZIG: Zigbee started"
✅	The autoexec.be file is present in the File Manager (≈45 KB)
✅	The Backlog command has been executed and the device has restarted
✅	The LumoHome app can connect (MQTT broker details entered in settings)

💡 If the system is not working correctly, verify that the Host, Port, TLS, username, and password are correctly entered in the Tasmota MQTT settings. The most common mistake is an incorrect Topic or disabled MQTT TLS.